Penguins Database - PCA Analysis

Diplomado de métodos matemáticos y estadísticos para la Ciencia de Datos

Equipo 7
Integrantes:
Fajardo Zavala Yendali Viridiana
González Rivas José Rafael
Maldonado Zamora Silvia Viridiana
Olivera Flores Jonathan
Valencia Quiroz Luis Roberto

2025-12-12

Descripción

La base de datos "penguins.csv" es un conjunto de datos sobre las mediciones biométricas de pingüinos en las Islas Palmer, Antártida. Contiene 344 registros y 8 variables.

Formato

Las variables incluidas son:

1. species: Categórica. La especie del pingüino (Adelie, Chinstrap o Gentoo).
2. island: Categórica. La isla de observación (Torgersen, Biscoe o Dream).
3. bill_length_mm: Numérica (float). Longitud del pico (culmen) en milímetros. Se observan 2 valores faltantes.
4. bill_depth_mm: Numérica (float). Profundidad del pico en milímetros. Se observan 2 valores faltantes.
5. flipper_length_mm: Numérica (float). Longitud de la aleta en milímetros. Se observan 2 valores faltantes.
6. body_mass_g: Numérica (float). Masa corporal en gramos. Se observan 2 valores faltantes.
7. sex: Categórica. El sexo del pingüino (male o female). Se observan 11 valores faltantes.
8. year: Numérica (integer). El año de la observación.

Carga de Datos

```r
df<-read.csv("penguins.csv")

df %>% dim()
```

```
## [1] 344   8
```

```r
df %>% glimpse()
```

```
## Rows: 344
## Columns: 8
## $ species           <chr> "Adelie", "Adelie", "Adelie", "Adelie", "Adelie", "A~
## $ island            <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
## $ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex               <chr> "male", "female", "female", NA, "female", "male", "f~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```r
df %>% head()
```

```
##   species    island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## 1  Adelie Torgersen           39.1          18.7               181        3750
## 2  Adelie Torgersen           39.5          17.4               186        3800
## 3  Adelie Torgersen           40.3          18.0               195        3250
## 4  Adelie Torgersen             NA            NA                NA          NA
## 5  Adelie Torgersen           36.7          19.3               193        3450
## 6  Adelie Torgersen           39.3          20.6               190        3650
##      sex year
## 1   male 2007
## 2 female 2007
## 3 female 2007
## 4   <NA> 2007
## 5 female 2007
## 6   male 2007
```

```r
###Valores faltantes

colSums(is.na(df))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 2                 2
## flipper_length_mm       body_mass_g               sex              year
##                 2                 2                11                 0
```

```r
###Variable de clasificacion: diagnosis

df <- df %>%
  mutate(species = as.factor(species))

df %>% count(species)
```

```
##     species   n
## 1    Adelie 152
## 2 Chinstrap  68
## 3    Gentoo 124
```

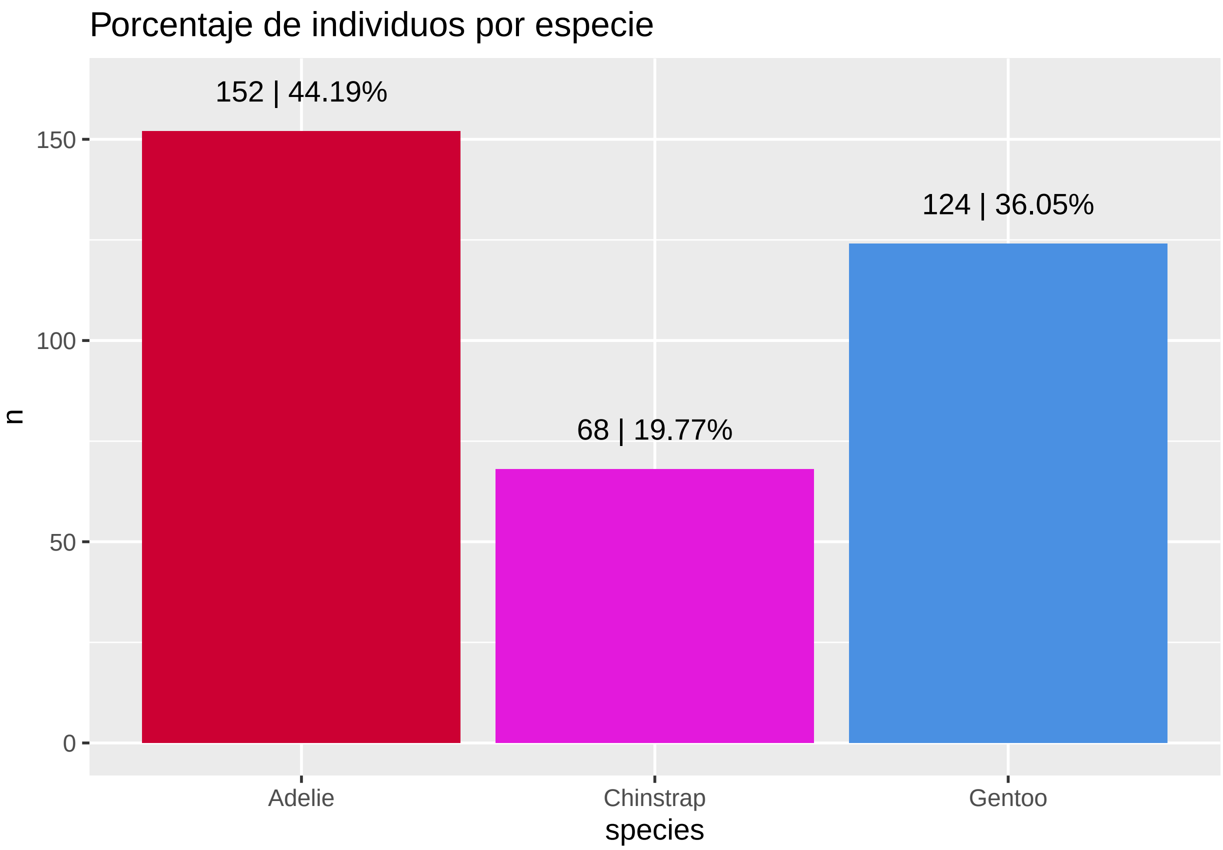
```r
table(df$species)
```

```
##
##    Adelie Chinstrap    Gentoo
##       152        68       124
```

```r
df %>% glimpse()
```

```
## Rows: 344
## Columns: 8
## $ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island            <chr> "Torgersen", "Torgersen", "Torgersen", "Torgersen", ~
## $ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
## $ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
## $ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
## $ sex               <chr> "male", "female", "female", NA, "female", "male", "f~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```r
df %>%
  group_by(species) %>%
  summarise(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = species, y = n)) +
    geom_col(fill = c("#CC0033", "#e319dc", "#4A90E2")) +
    geom_text(aes(label = paste0(n, " | ",
                                 signif(n / nrow(df) * 100, digits = 4), '%')),
              nudge_y = 10) +
    ggtitle("Porcentaje de individuos por especie") +
    theme_gray()
```



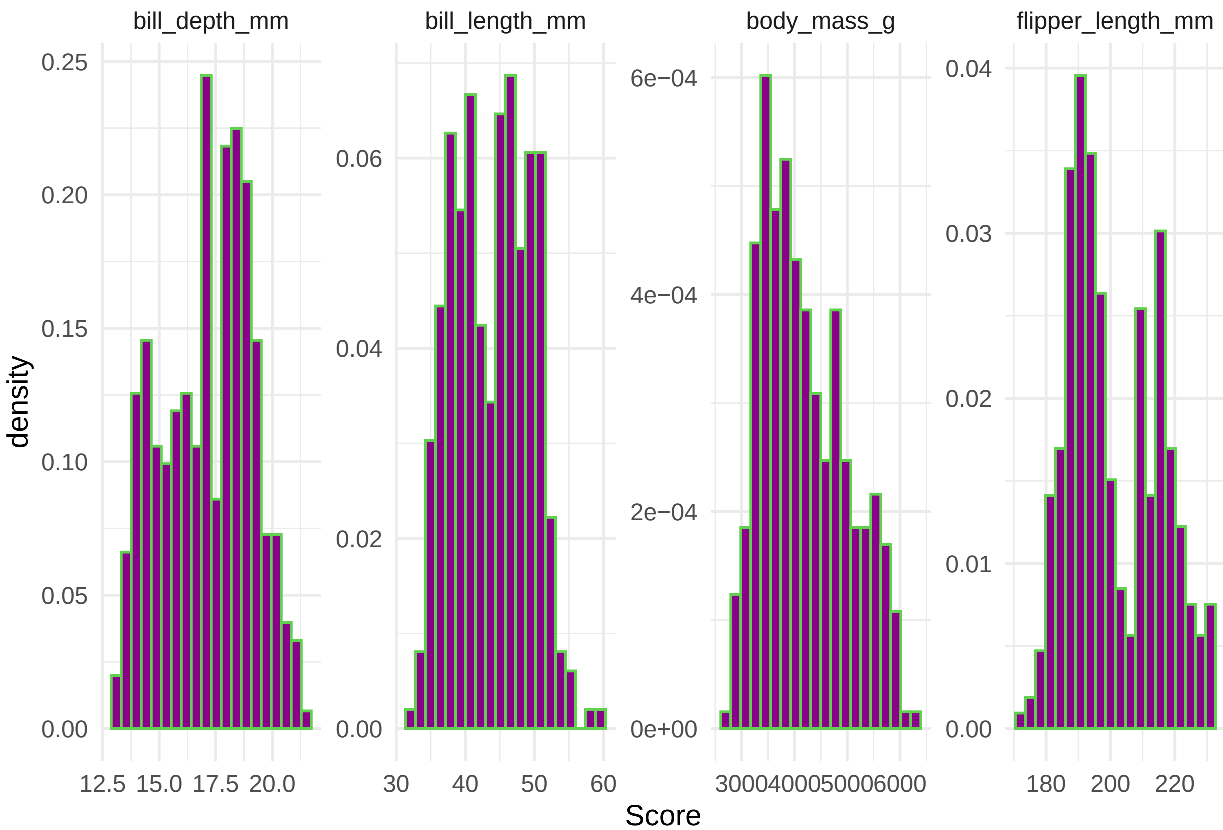

```r
### Selección numérica correcta

df1 <- df %>%
  select(where(is.numeric)) %>%  # solo numéricas
  select(-year) %>%              # EXCLUYE year
  drop_na()                      # elimina NA

df1 %>% head()
```

```
##   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## 1           39.1          18.7               181        3750
## 2           39.5          17.4               186        3800
## 3           40.3          18.0               195        3250
## 4           36.7          19.3               193        3450
## 5           39.3          20.6               190        3650
## 6           38.9          17.8               181        3625
```

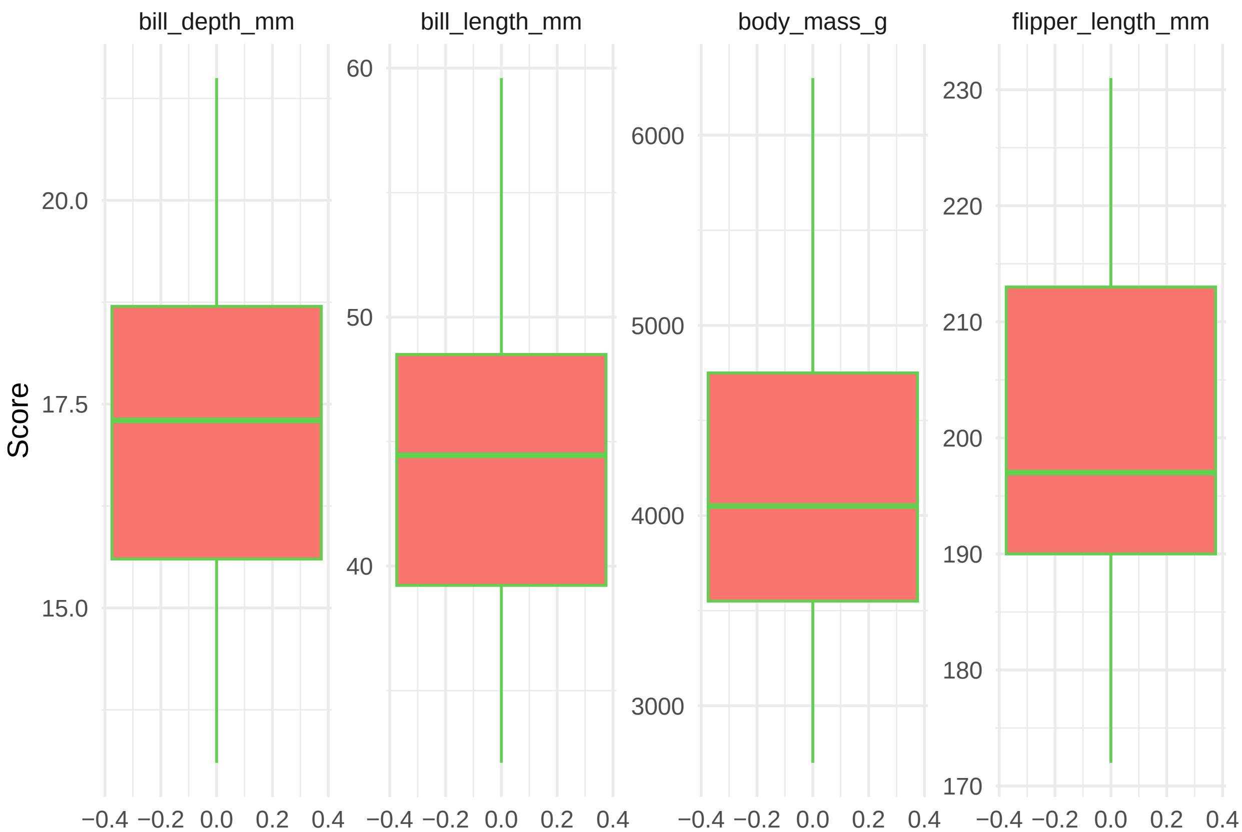
```r
### Histogramas

df1 |> pivot_longer(everything(),
                    names_to = "Variable",
                    values_to = "Score") |>
  ggplot(aes(x = Score)) +
  geom_histogram(aes(y = ..density..),
                 bins = 20, colour = 3,
                 fill = "darkmagenta") +
```

```
  facet_wrap("Variable", ncol = 4, scales = "free") +
  theme_minimal()
```

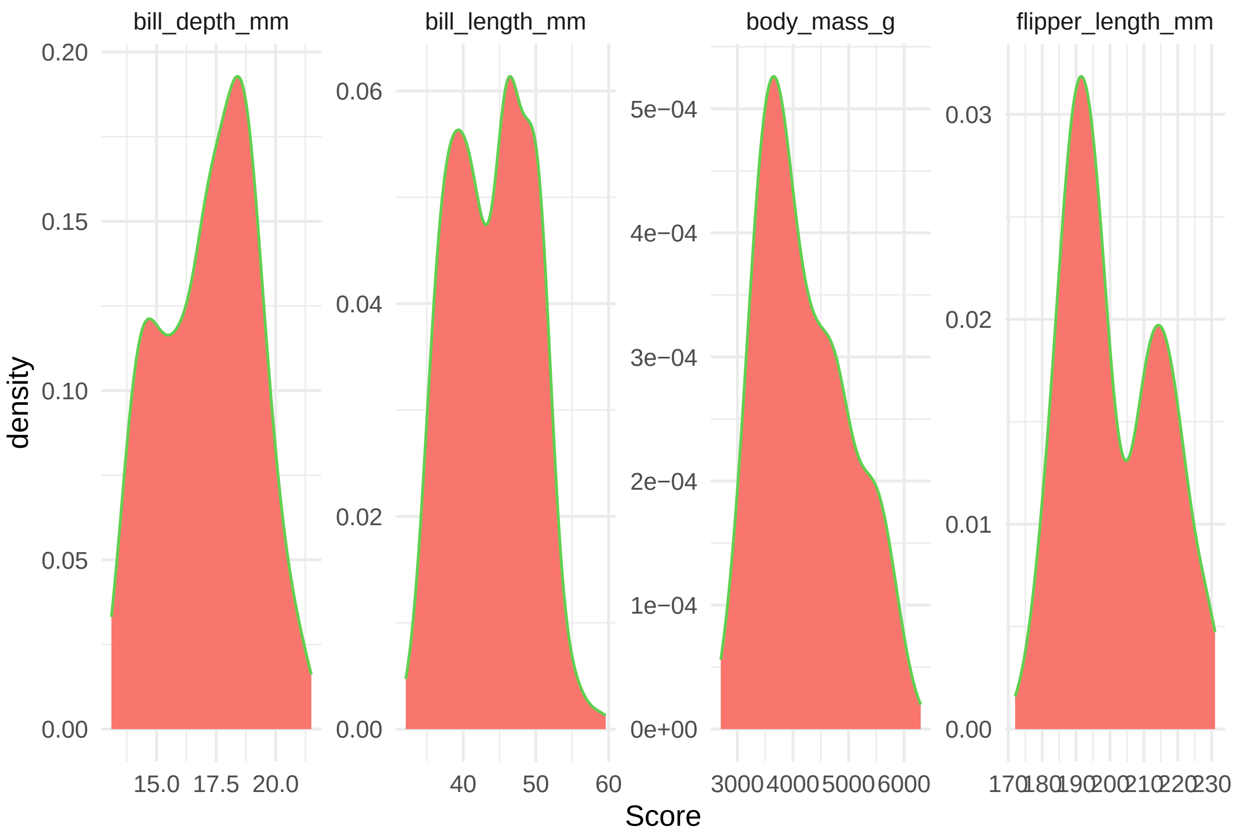
```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



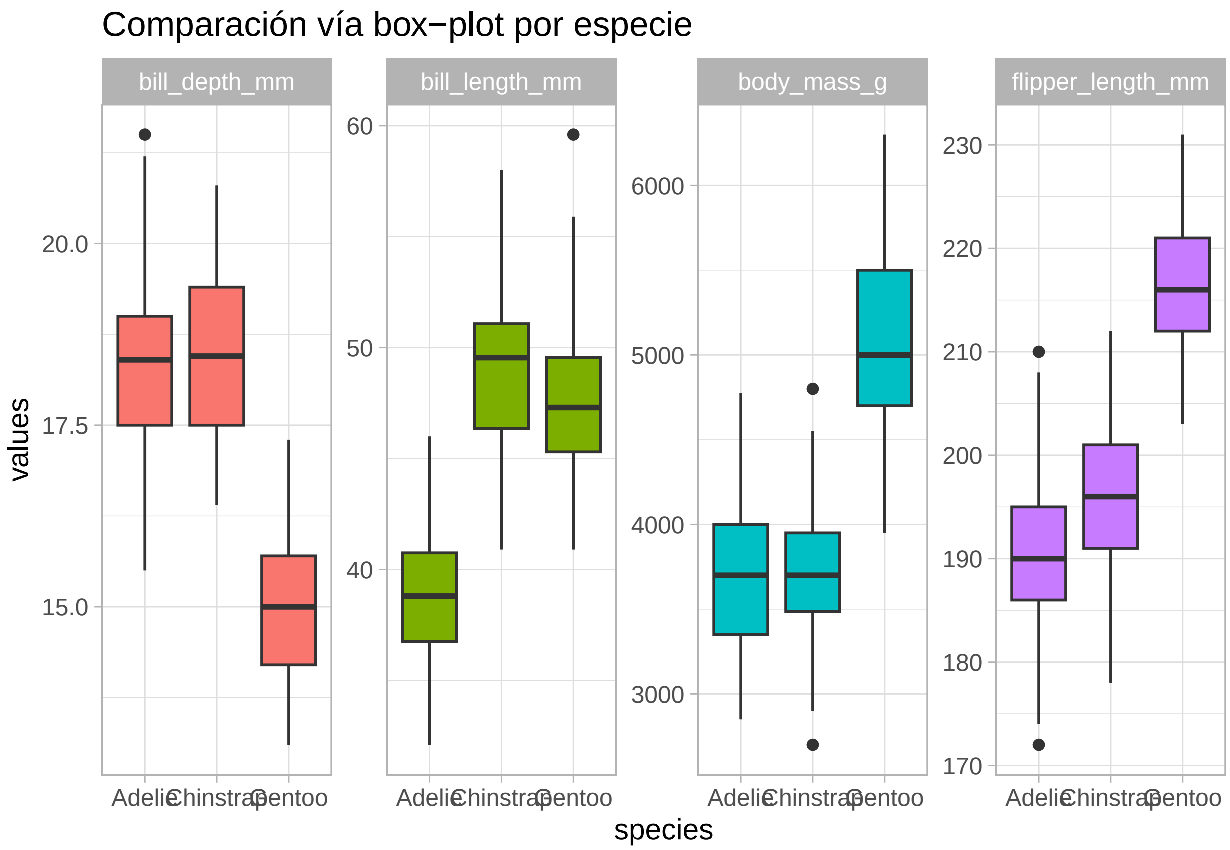

```
### Boxplots

df1 |> pivot_longer(everything(),
                    names_to = "Variable",
                    values_to = "Score") |>
  ggplot(aes(y = Score)) +
  geom_boxplot(aes(fill = "darkred"),
               colour = 3, show.legend = FALSE) +
  facet_wrap("Variable", ncol = 4, scales = "free") +
  theme_minimal()
```



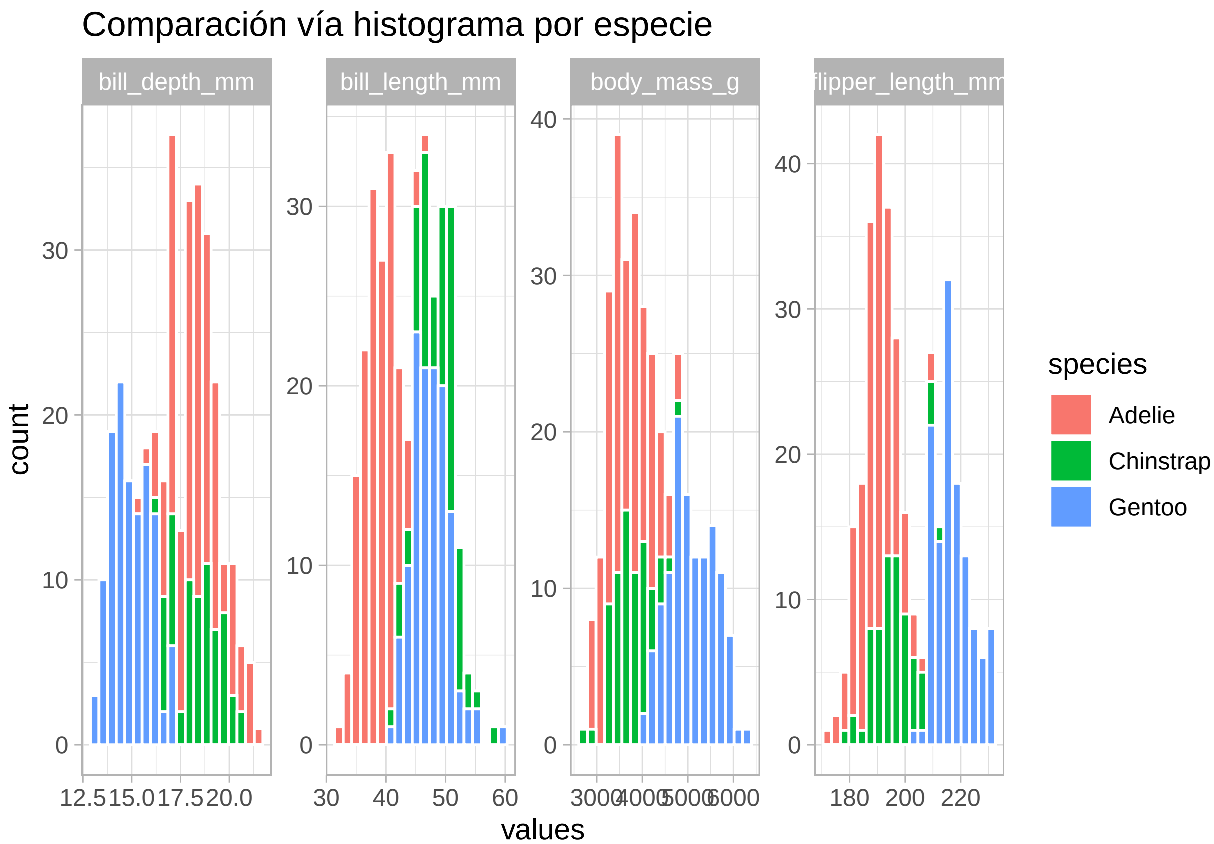

```
###Densidad

df1 |> pivot_longer(everything(),
   names_to = "Variable",values_to="Score") |>
     ggplot(aes(x=Score)) + geom_density(aes(fill="darkred"),colour = 3,show.legend = FALSE) +
       facet_wrap("Variable",ncol = 4,scales = "free" ) + theme_minimal()
```



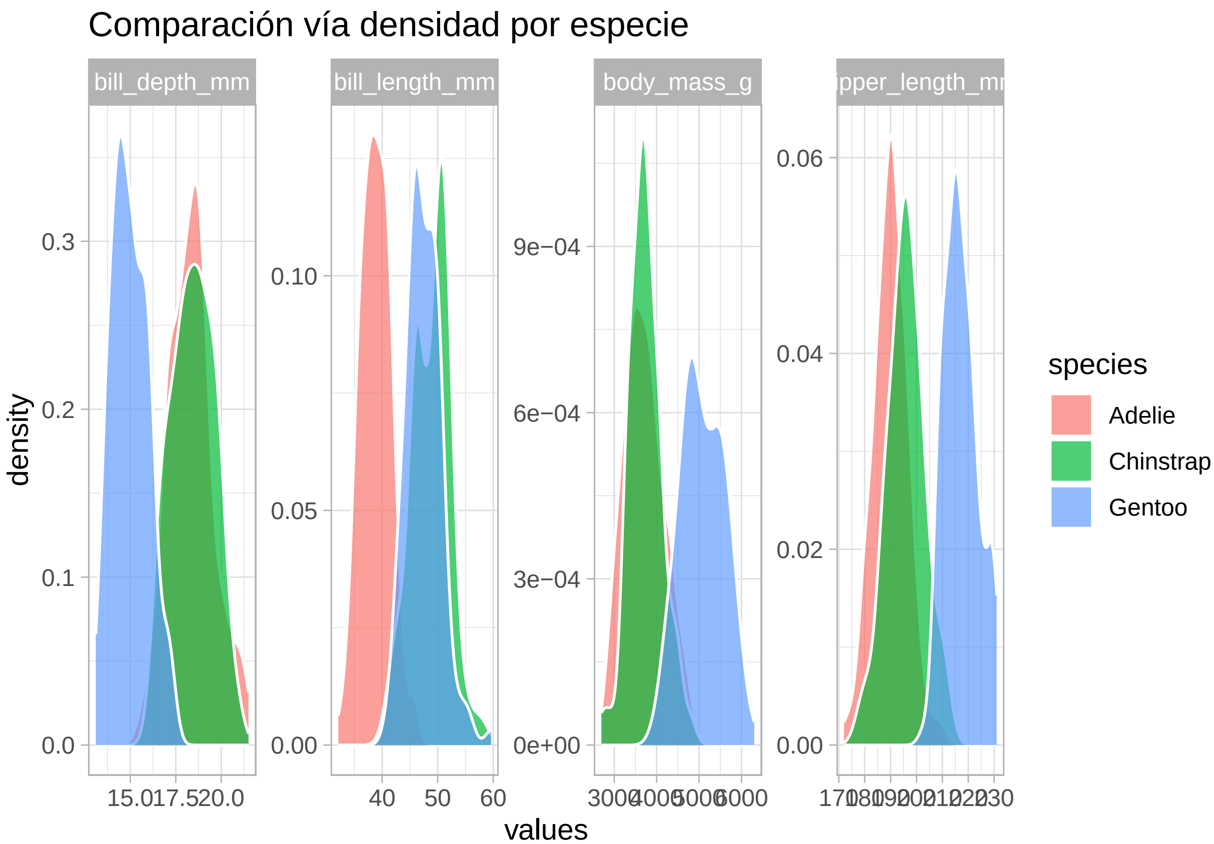

```r
df1_con_clase <- df %>%
  select(species, where(is.numeric)) %>%
  select(-year) %>%
  drop_na()


### Comparación por la variable de clasificación: species

df1_long <- df1_con_clase %>%
  pivot_longer(!species,
               names_to = "predictores",
               values_to = "values")

theme_set(theme_light())

df1_long %>%
  ggplot(aes(x = species, y = values, fill = predictores)) +
    geom_boxplot() +
    facet_wrap(~ predictores, scales = "free", ncol = 4) +
    scale_color_viridis_d(option = "plasma", end = .7) +
    theme(legend.position = "none") +
    labs(title = "Comparación vía box-plot por especie")
```



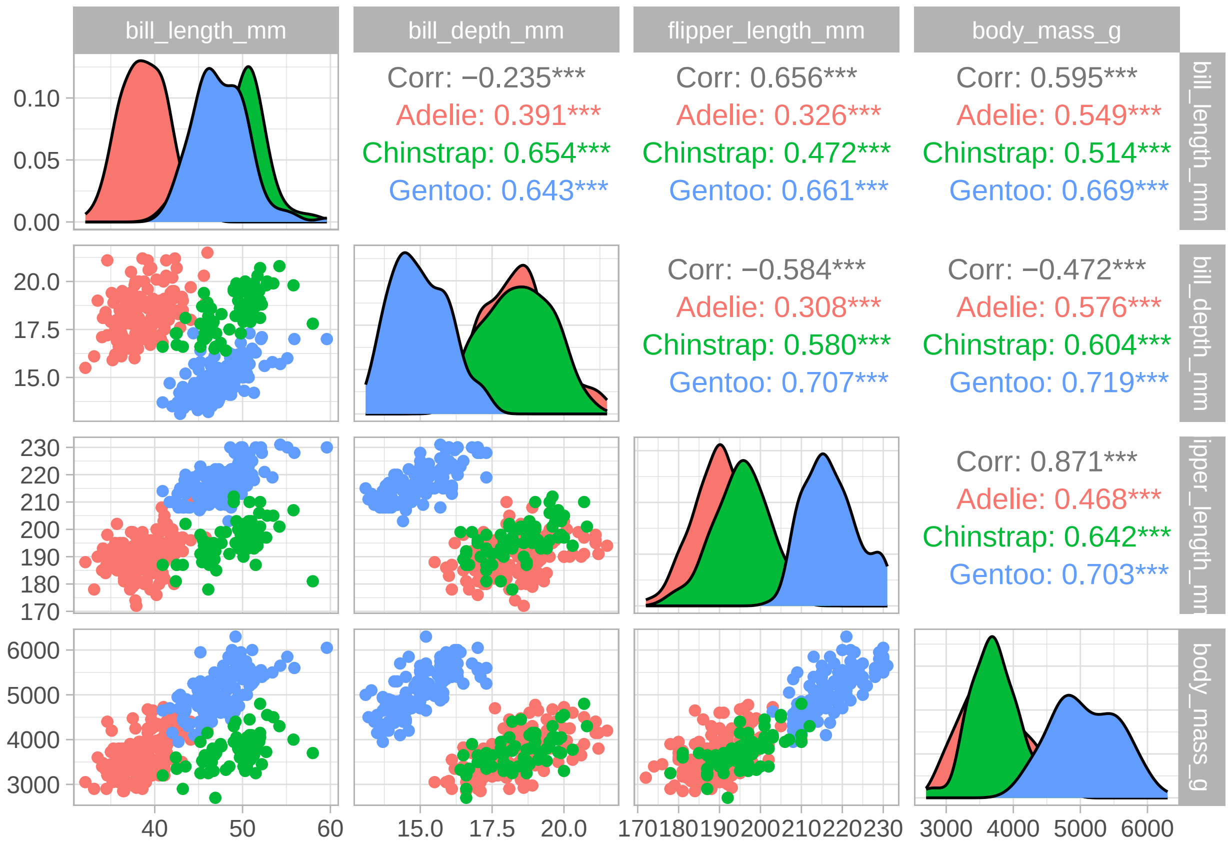

```r
df1_long %>%
  ggplot(aes(values, fill = species)) +
    geom_histogram(color = "white", bins = 20) +
    facet_wrap(~ predictores, scales = "free", ncol = 4) +
    scale_color_viridis_d(option = "plasma", end = .7) +
    theme_light() +
    labs(title = "Comparación vía histograma por especie")
```




```r
df1_long %>%
  ggplot(aes(values, fill = species)) +
    geom_density(color = "white", alpha = 0.7) +
    facet_wrap(~ predictores, scales = "free", ncol = 4) +
    scale_color_viridis_d(option = "plasma", end = .7) +
    theme_light() +
    labs(title = "Comparación vía densidad por especie")
```




```
ggpairs(df1_con_clase,
        mapping = aes(color = species),
        columns = 2:ncol(df1_con_clase))
```




```r
### Correlación
cor_peng <- df1 %>% correlate(method = "spearman")
```

```
## Correlation computed with
## * Method: 'spearman'
## * Missing treated using: 'pairwise.complete.obs'
```

```r
cor_peng
```

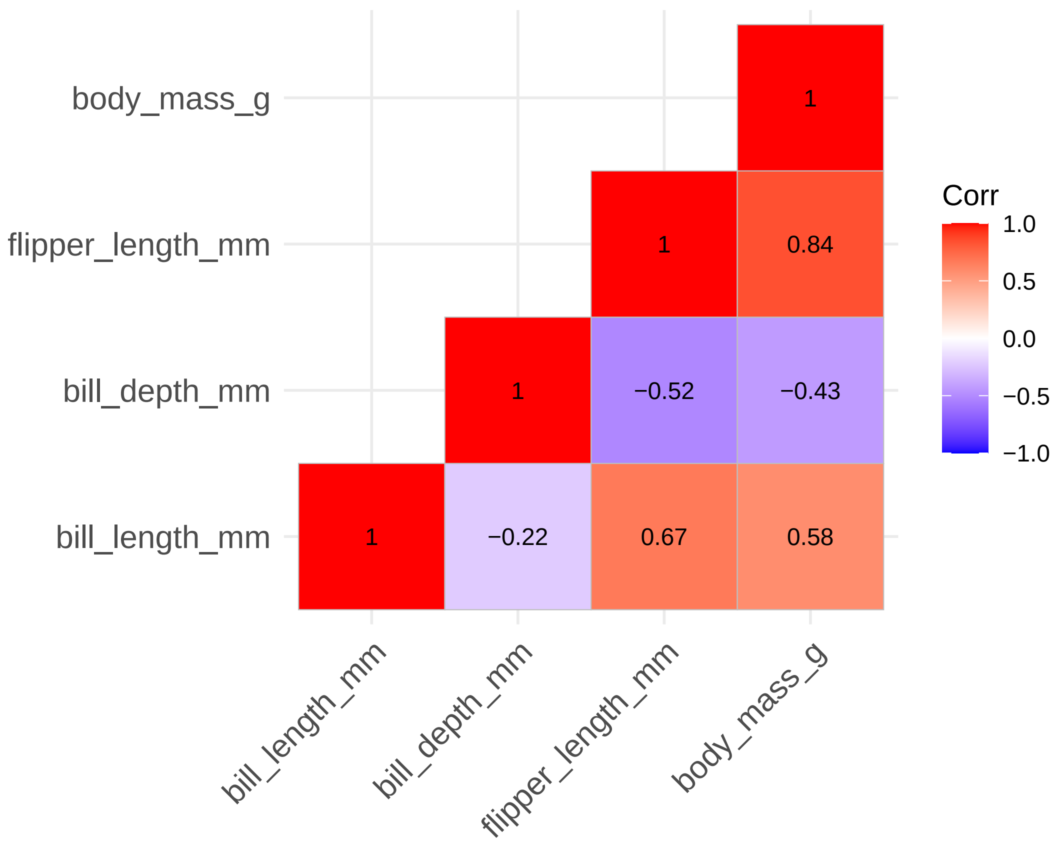
```
## # A tibble: 4 x 5
##   term              bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>                      <dbl>         <dbl>             <dbl>       <dbl>
## 1 bill_length_mm            NA            -0.222             0.673       0.584
## 2 bill_depth_mm             -0.222        NA                -0.523      -0.432
## 3 flipper_length_mm          0.673        -0.523            NA           0.840
## 4 body_mass_g                0.584        -0.432             0.840      NA
```

```r
cor_matrix <- cor(df1, method = "spearman")
cor_matrix
```

```
##                   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## bill_length_mm         1.0000000    -0.2217492         0.6727719   0.5838003
## bill_depth_mm         -0.2217492     1.0000000        -0.5232675  -0.4323722
## flipper_length_mm      0.6727719    -0.5232675         1.0000000   0.8399741
## body_mass_g            0.5838003    -0.4323722         0.8399741   1.0000000
```

```r
ggcorrplot::ggcorrplot(corr = cor_matrix,
                       type = "lower",
                       show.diag = TRUE,
                       lab = TRUE,
                       lab_size = 3)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the ggcorrplot package.
##   Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
det(cor_matrix)
```

```
## [1] 0.1117964
```

```r
psych::KMO(cor_matrix)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: psych::KMO(r = cor_matrix)
## Overall MSA =  0.7
```

```
## MSA for each item =
##    bill_length_mm     bill_depth_mm flipper_length_mm       body_mass_g
##              0.77              0.73              0.63              0.72
```

```r
###Prueba de Bartlett

psych::cortest.bartlett(cor_matrix, n = nrow(df1))
```

```
## $chisq
## [1] 742.4096
##
## $p.value
## [1] 4.249454e-157
##
## $df
## [1] 6
```

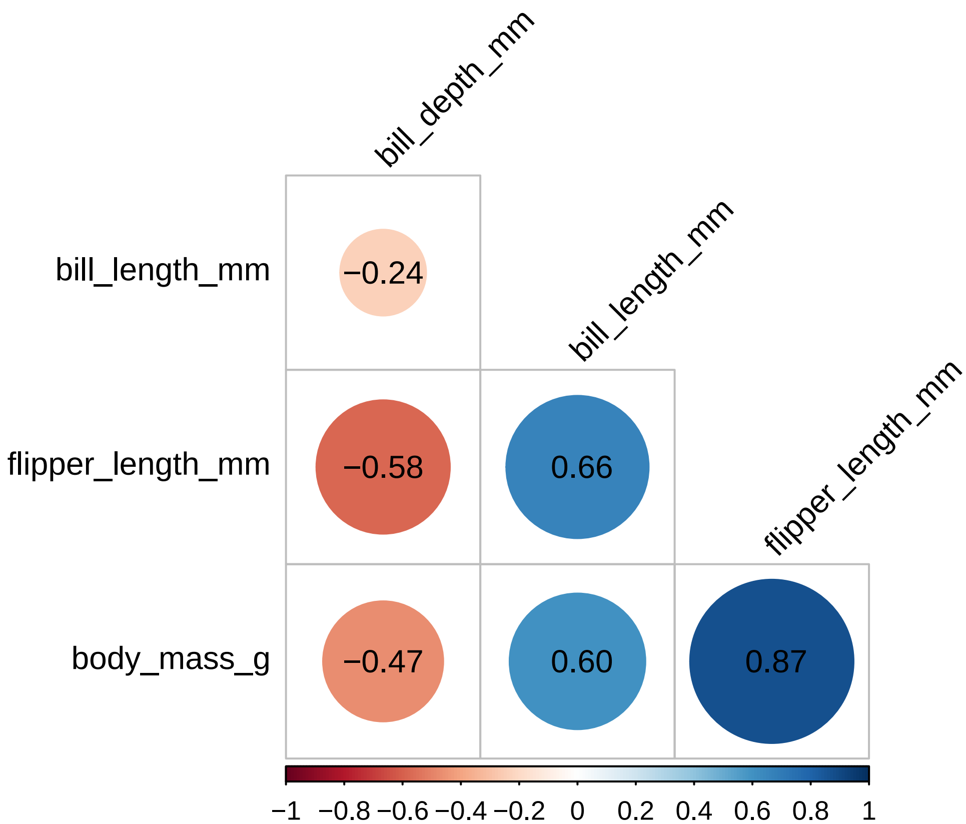
```r
cor.df1 <- df1 %>% cor_mat()
cor.df1
```

```
## # A tibble: 4 x 5
##   rowname           bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## * <chr>                      <dbl>         <dbl>             <dbl>       <dbl>
## 1 bill_length_mm              1            -0.24              0.66        0.6
## 2 bill_depth_mm              -0.24          1                -0.58       -0.47
## 3 flipper_length_mm           0.66         -0.58              1           0.87
## 4 body_mass_g                 0.6          -0.47              0.87        1
```

```r
options(scipen = 999)

### p-values
cor.df1 %>% cor_get_pval()
```

```
## # A tibble: 4 x 5
##   rowname           bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <chr>                      <dbl>         <dbl>             <dbl>       <dbl>
## 1 bill_length_mm          0             1.12e- 5         1.74e- 43   3.81e- 34
## 2 bill_depth_mm           1.12e- 5      0                1.23e- 32   2.28e- 20
## 3 flipper_length_mm       1.74e-43      1.23e-32         0           4.37e-107
## 4 body_mass_g             3.81e-34      2.28e-20         4.37e-107   0
```

```r
### Graficar matriz reordenada
cor.df1 %>%
  cor_reorder() %>%
  pull_lower_triangle() %>%
  cor_plot(label = TRUE)
```




```r
### Mostrar todos los pares de correlaciones
cor.df1 %>% cor_gather() %>% print(n = Inf)
```

```
## # A tibble: 16 x 4
##    var1              var2                cor         p
##    <chr>             <chr>             <dbl>     <dbl>
##  1 bill_length_mm    bill_length_mm     1     0
##  2 bill_depth_mm     bill_length_mm    -0.24 1.12e-  5
##  3 flipper_length_mm bill_length_mm     0.66 1.74e- 43
##  4 body_mass_g       bill_length_mm     0.6  3.81e- 34
##  5 bill_length_mm    bill_depth_mm     -0.24 1.12e-  5
##  6 bill_depth_mm     bill_depth_mm      1     0
##  7 flipper_length_mm bill_depth_mm     -0.58 1.23e- 32
##  8 body_mass_g       bill_depth_mm     -0.47 2.28e- 20
##  9 bill_length_mm    flipper_length_mm  0.66 1.74e- 43
## 10 bill_depth_mm     flipper_length_mm -0.58 1.23e- 32
## 11 flipper_length_mm flipper_length_mm  1     0
## 12 body_mass_g       flipper_length_mm  0.87 4.37e-107
## 13 bill_length_mm    body_mass_g        0.6  3.81e- 34
## 14 bill_depth_mm     body_mass_g       -0.47 2.28e- 20
## 15 flipper_length_mm body_mass_g        0.87 4.37e-107
## 16 body_mass_g       body_mass_g        1     0
```

```r
peng_recipe <-
  recipe(~ ., data = df1_con_clase) %>%
  update_role(species, new_role = "id") %>%
  step_dummy(all_nominal_predictors()) %>%
```

```r
  step_normalize(all_predictors()) %>%
  step_pca(all_predictors(), id = "pca")
  # %>% step_pca(all_numeric(), threshold = 0.8)

peng_recipe
```

```
##
## -- Recipe ----------------------------------------------------------------------
##
## -- Inputs
## Number of variables by role
## predictor: 4
## id:        1
##
## -- Operations
## * Dummy variables from: all_nominal_predictors()
## * Centering and scaling for: all_predictors()
## * PCA extraction with: all_predictors()
```

```r
### Preparar el PCA

peng_pca <- prep(peng_recipe)

### Cargas de las variables en los componentes
pca_loading <- tidy(peng_pca, id = "pca")
pca_loading
```

```
## # A tibble: 16 x 4
##    terms                value component id
##    <chr>                <dbl> <chr>     <chr>
##  1 bill_length_mm     0.455   PC1       pca
##  2 bill_depth_mm     -0.400   PC1       pca
##  3 flipper_length_mm  0.576   PC1       pca
##  4 body_mass_g        0.548   PC1       pca
##  5 bill_length_mm    -0.597   PC2       pca
##  6 bill_depth_mm     -0.798   PC2       pca
##  7 flipper_length_mm -0.00228 PC2       pca
##  8 body_mass_g       -0.0844  PC2       pca
##  9 bill_length_mm    -0.644   PC3       pca
## 10 bill_depth_mm      0.418   PC3       pca
## 11 flipper_length_mm  0.232   PC3       pca
## 12 body_mass_g        0.597   PC3       pca
## 13 bill_length_mm     0.146   PC4       pca
## 14 bill_depth_mm     -0.168   PC4       pca
## 15 flipper_length_mm -0.784   PC4       pca
## 16 body_mass_g        0.580   PC4       pca
```

```r
### Varianza explicada por cada componente
pca_variances <- tidy(peng_pca, id = "pca", type = "variance")
pca_variances
```

```
## # A tibble: 16 x 4
```

```
##    terms                          value component id
##    <chr>                          <dbl>     <int> <chr>
##  1 variance                       2.75          1 pca
##  2 variance                       0.773         2 pca
##  3 variance                       0.365         3 pca
##  4 variance                       0.108         4 pca
##  5 cumulative variance            2.75          1 pca
##  6 cumulative variance            3.53          2 pca
##  7 cumulative variance            3.89          3 pca
##  8 cumulative variance            4.00          4 pca
##  9 percent variance              68.8           1 pca
## 10 percent variance              19.3           2 pca
## 11 percent variance               9.13          3 pca
## 12 percent variance               2.71          4 pca
## 13 cumulative percent variance  68.8           1 pca
## 14 cumulative percent variance  88.2           2 pca
## 15 cumulative percent variance  97.3           3 pca
## 16 cumulative percent variance 100             4 pca
```

```
### Porcentaje de varianza explicado por cada PC
pca_var_percent <- tidy(peng_pca, id = "pca", type = "variance") %>%
  filter(str_detect(terms, "percent variance"))
pca_var_percent
```

```
## # A tibble: 8 x 4
##   terms                        value component id
##   <chr>                        <dbl>     <int> <chr>
## 1 percent variance             68.8          1 pca
## 2 percent variance             19.3          2 pca
## 3 percent variance              9.13         3 pca
## 4 percent variance              2.71         4 pca
## 5 cumulative percent variance  68.8          1 pca
## 6 cumulative percent variance  88.2          2 pca
## 7 cumulative percent variance  97.3          3 pca
## 8 cumulative percent variance 100            4 pca
```

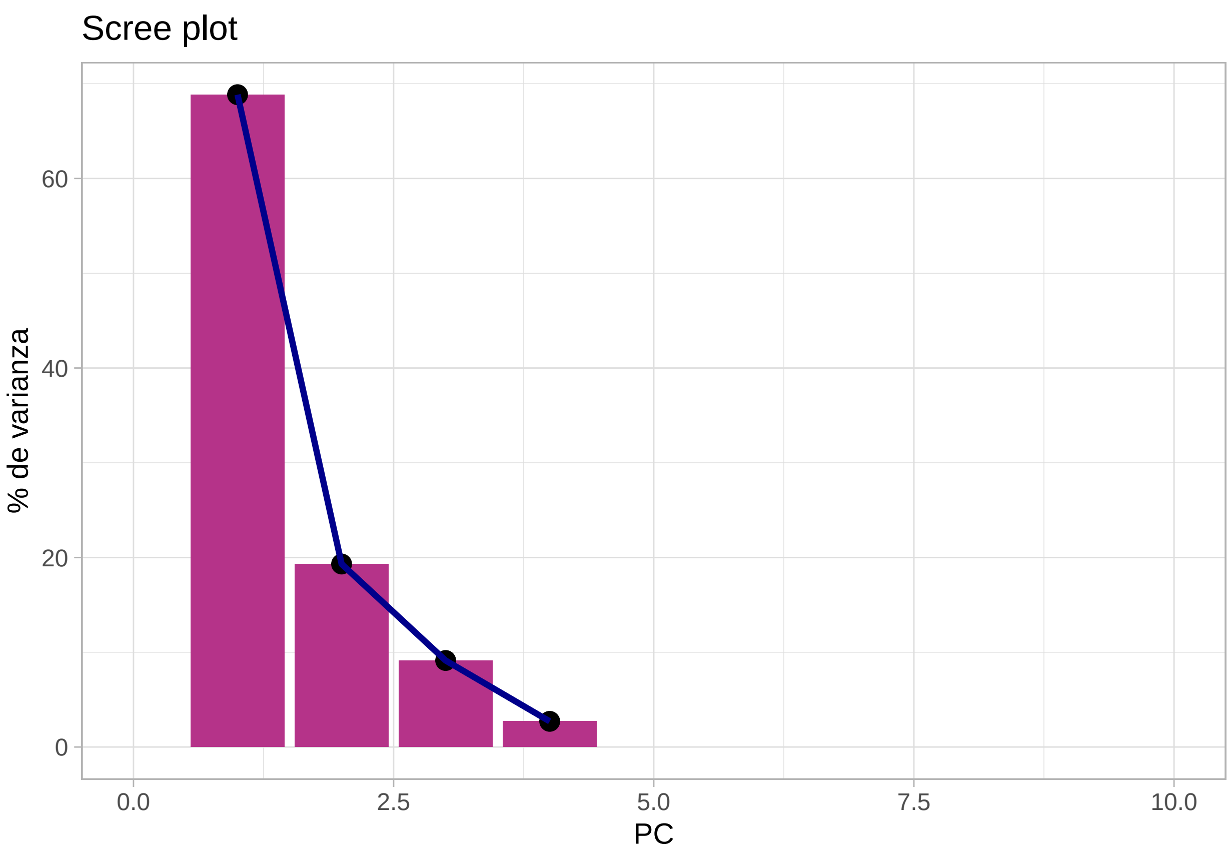
```
### Varianza acumulada
pca_var_cum_percent <- tidy(peng_pca, id = "pca", type = "variance") %>%
  filter(str_detect(terms, "cumulative percent variance"))
pca_var_cum_percent
```

```
## # A tibble: 4 x 4
##   terms                       value component id
##   <chr>                       <dbl>     <int> <chr>
## 1 cumulative percent variance  68.8         1 pca
## 2 cumulative percent variance  88.2         2 pca
## 3 cumulative percent variance  97.3         3 pca
## 4 cumulative percent variance 100           4 pca
```

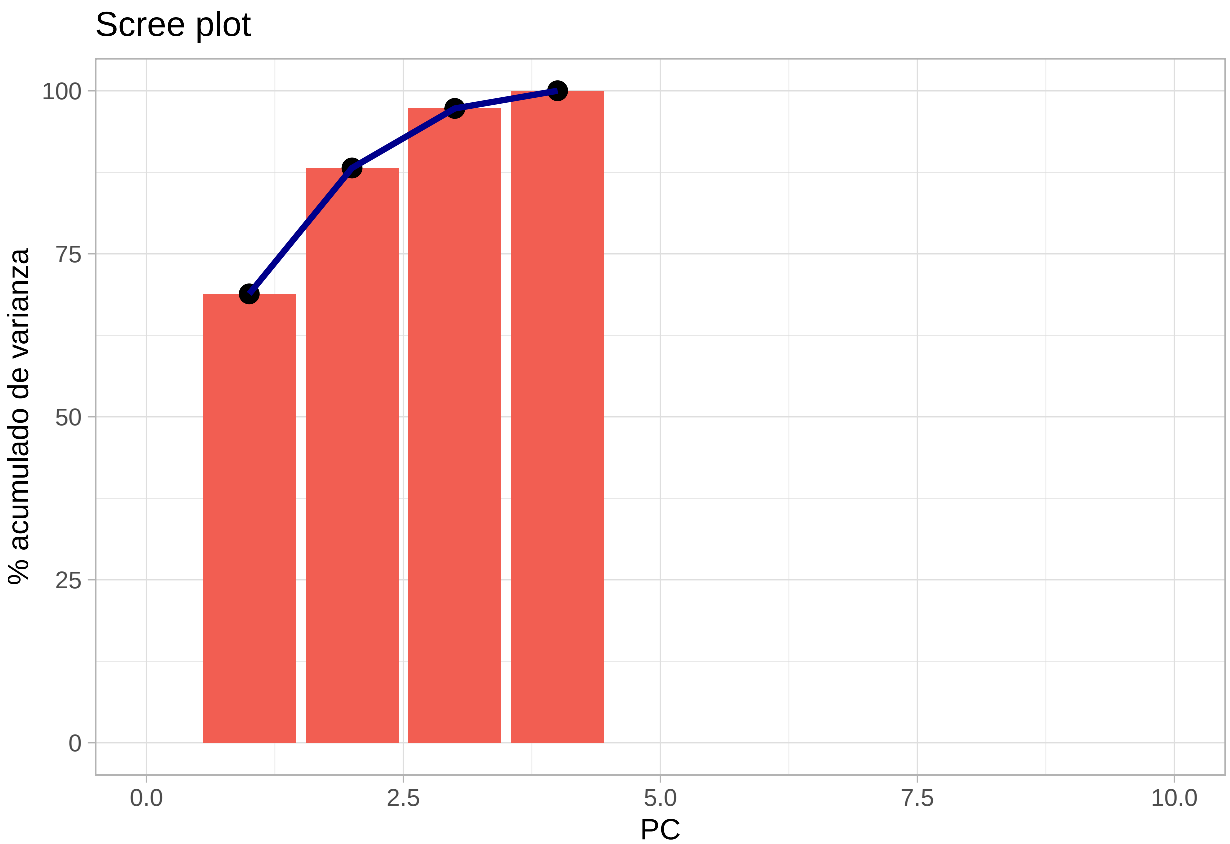
```
peng_pca %>%
  tidy(id = "pca", type = "variance") %>%
  dplyr::filter(terms == "percent variance") %>%
```

```r
  ggplot(aes(x = component, y = value)) +
  geom_col(fill = "#B53389") +
  xlim(c(0, 10)) +
  geom_point(size=3) +
  geom_line(color="darkblue", size=1.1)+
  labs(x="PC", y="% de varianza", title="Scree plot")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
peng_pca %>%
  tidy(id = "pca", type = "variance") %>%
  dplyr::filter(terms == "cumulative percent variance") %>%
  ggplot(aes(x = component, y = value)) +
  geom_col(fill = "#F25E52") +
  xlim(c(0, 10)) +
  geom_point(size=3) +
  geom_line(color="darkblue", size=1.1)+
  labs(x="PC", y="% acumulado de varianza", title="Scree plot")
```




```r
variance_exp <- tidy(peng_pca,id = "pca", type = "variance") %>%
                filter(str_detect(terms, "percent variance"))

variance_exp
```

```
## # A tibble: 8 x 4
##   terms                         value component id
##   <chr>                         <dbl>     <int> <chr>
## 1 percent variance              68.8          1 pca
## 2 percent variance              19.3          2 pca
## 3 percent variance               9.13         3 pca
## 4 percent variance               2.71         4 pca
## 5 cumulative percent variance  68.8          1 pca
## 6 cumulative percent variance  88.2          2 pca
## 7 cumulative percent variance  97.3          3 pca
## 8 cumulative percent variance 100            4 pca
```

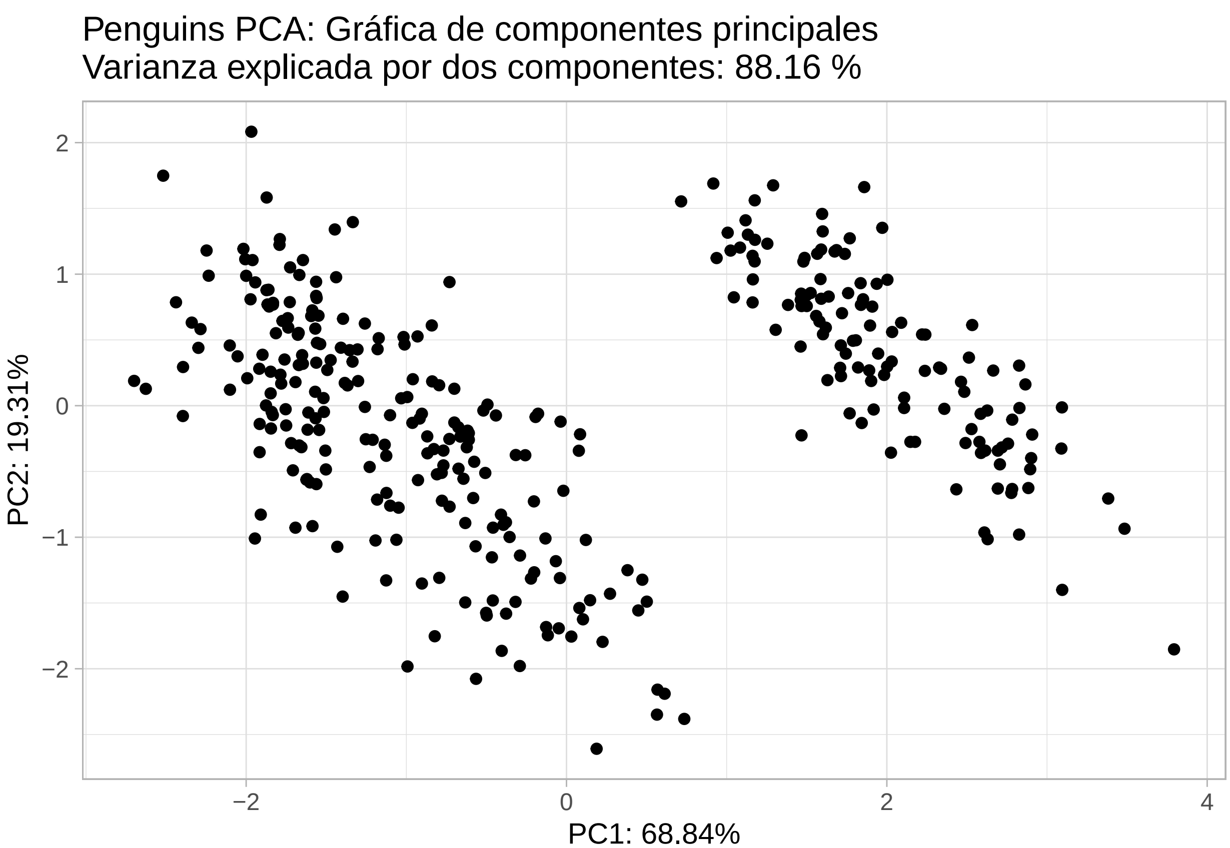
```r
###Grafica de los primeros dos componentes

VE <- paste(
  "Varianza explicada por dos componentes:",
  round(variance_exp$value[[1]] + variance_exp$value[[2]], digits = 2), "%"
)
VE
```

```
## [1] "Varianza explicada por dos componentes: 88.16 %"
```

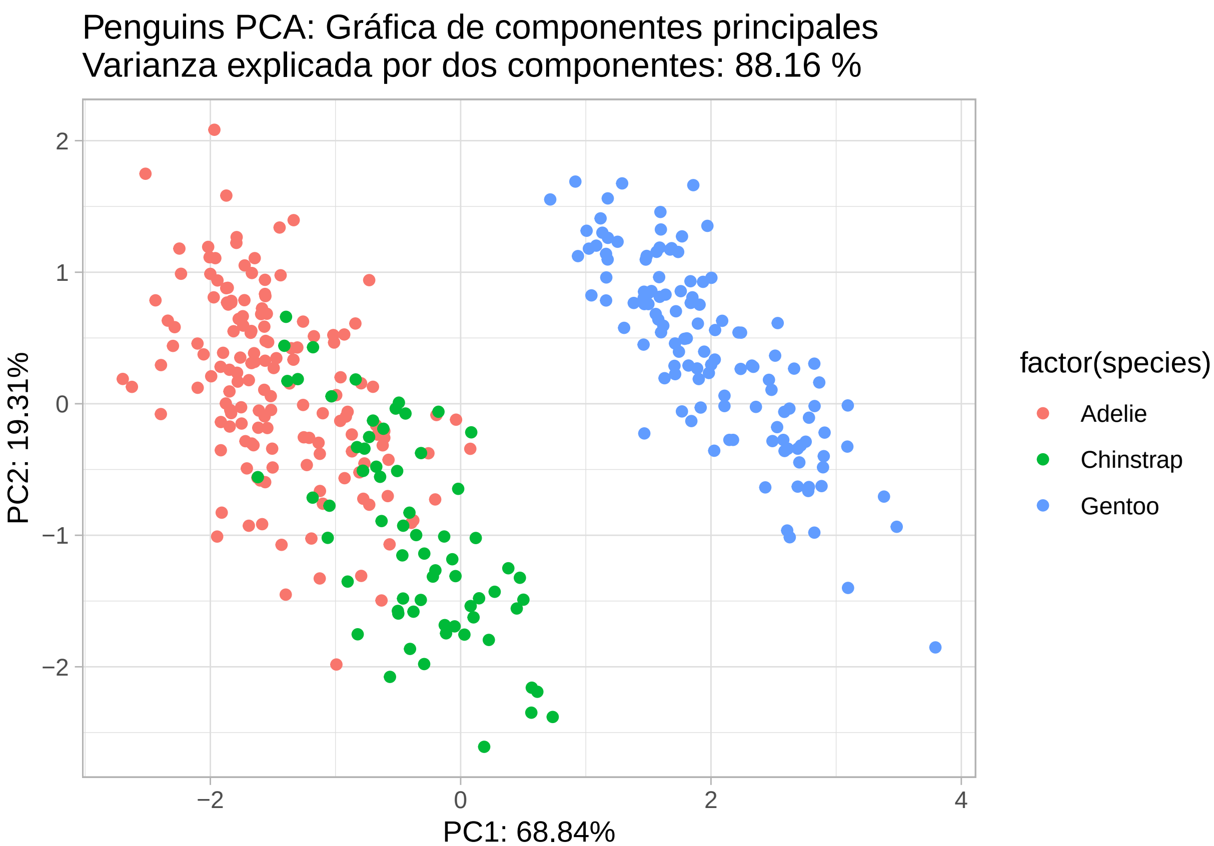
```r
datos_pca <- bake(peng_pca, new_data = NULL)
datos_pca
```

```
## # A tibble: 342 x 5
##    species   PC1      PC2     PC3     PC4
##    <fct>   <dbl>    <dbl>   <dbl>   <dbl>
##  1 Adelie -1.84  -0.0476   0.232   0.523
##  2 Adelie -1.30   0.428    0.0295  0.402
##  3 Adelie -1.37   0.154   -0.198  -0.527
##  4 Adelie -1.88   0.00205  0.618  -0.478
##  5 Adelie -1.91  -0.828    0.686  -0.207
##  6 Adelie -1.76   0.351   -0.0276  0.504
##  7 Adelie -0.809 -0.522    1.33    0.338
##  8 Adelie -1.83   0.769    0.689  -0.427
##  9 Adelie -1.19  -1.02     0.729   0.333
## 10 Adelie -1.73   0.787   -0.205   0.0205
## # i 332 more rows
```

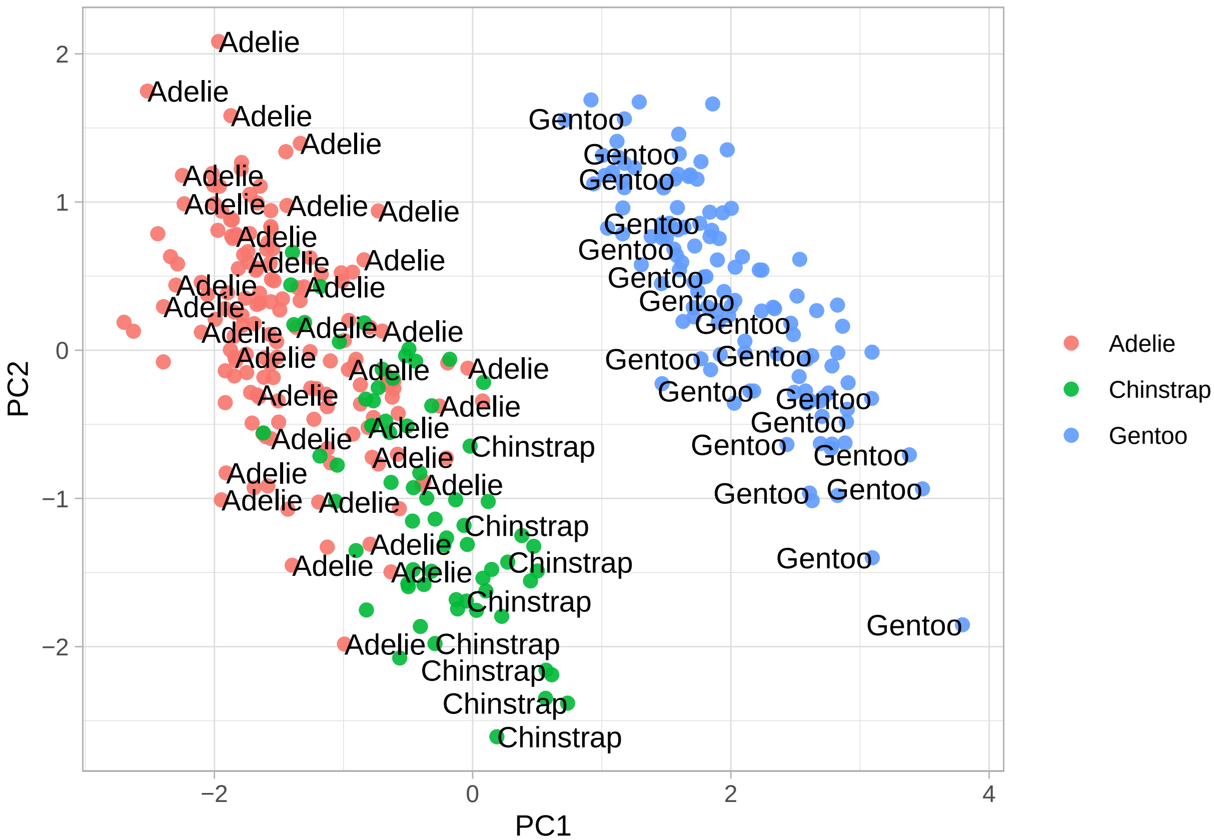
```r
datos_pca %>%
  ggplot(aes(x = PC1, y = PC2)) +
  geom_point() +
  labs(
    x = paste0("PC1: ", round(variance_exp$value[[1]], 2), "%"),
    y = paste0("PC2: ", round(variance_exp$value[[2]], 2), "%")
  ) +
  ggtitle(paste0("Penguins PCA: Gráfica de componentes principales", "\n", VE))
```



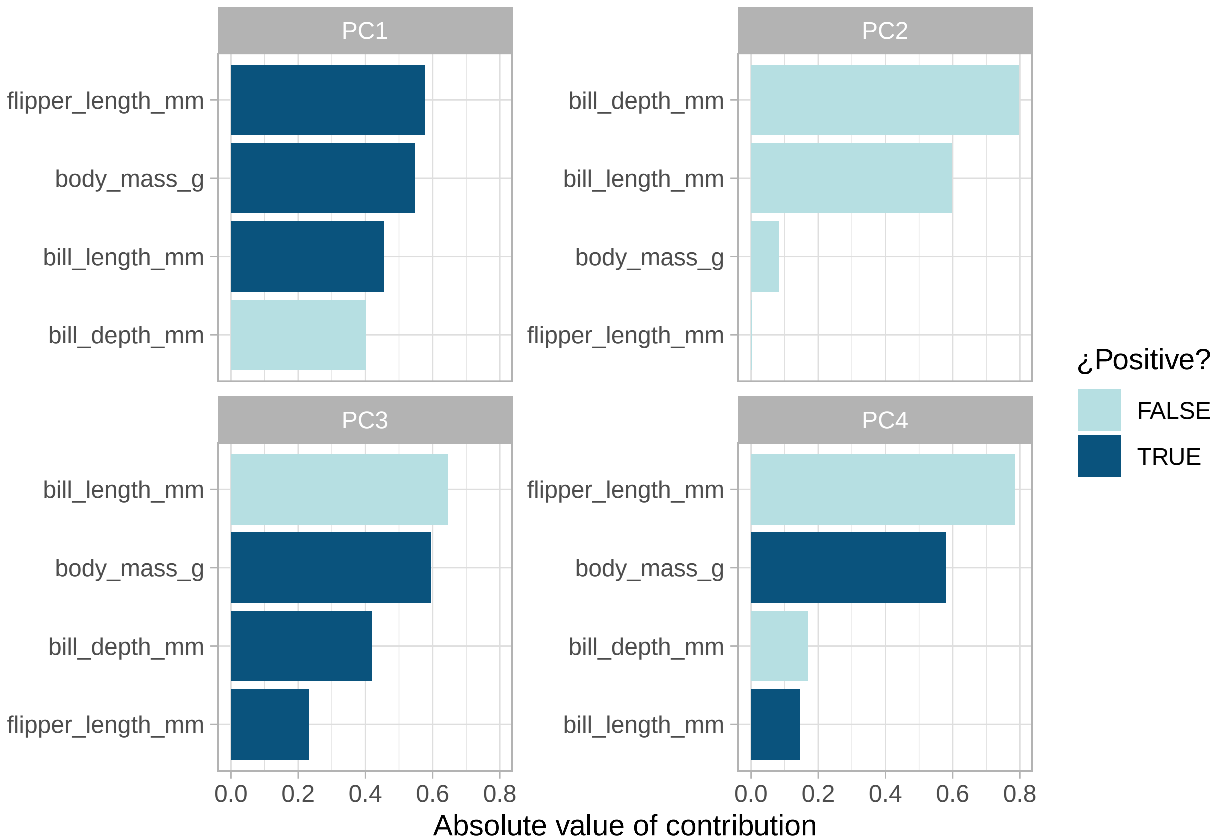
```r
datos_pca %>%
  ggplot(aes(x = PC1, y = PC2, color = factor(species))) +
  geom_point() +
  labs(
    x = paste0("PC1: ", round(variance_exp$value[[1]], 2), "%"),
    y = paste0("PC2: ", round(variance_exp$value[[2]], 2), "%")
  ) +
  ggtitle(paste0("Penguins PCA: Gráfica de componentes principales", "\n", VE))
```



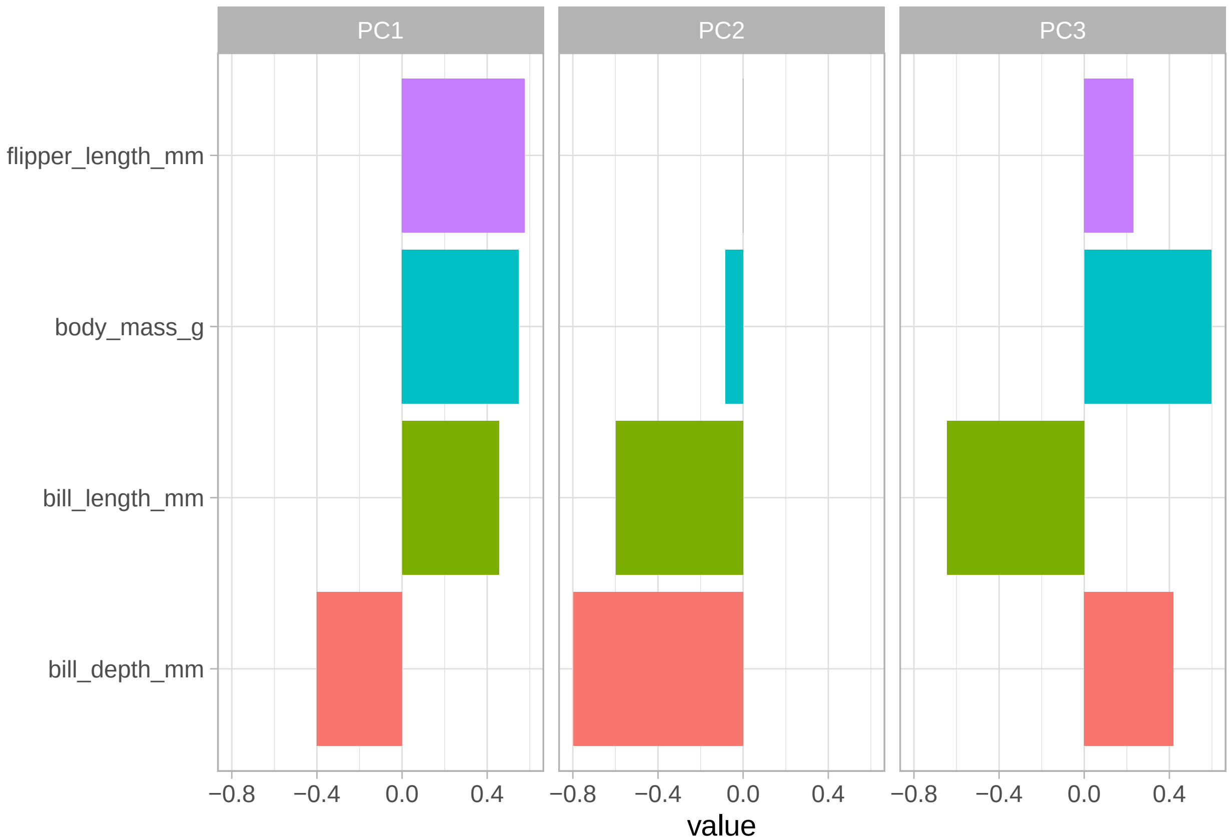

```r
juice(peng_pca) %>%
  ggplot(aes(PC1, PC2, label = species)) +
  geom_point(aes(color = species), alpha = 0.9, size = 2) +
  geom_text(check_overlap = TRUE, hjust = "inward") +
  labs(color = NULL)
```



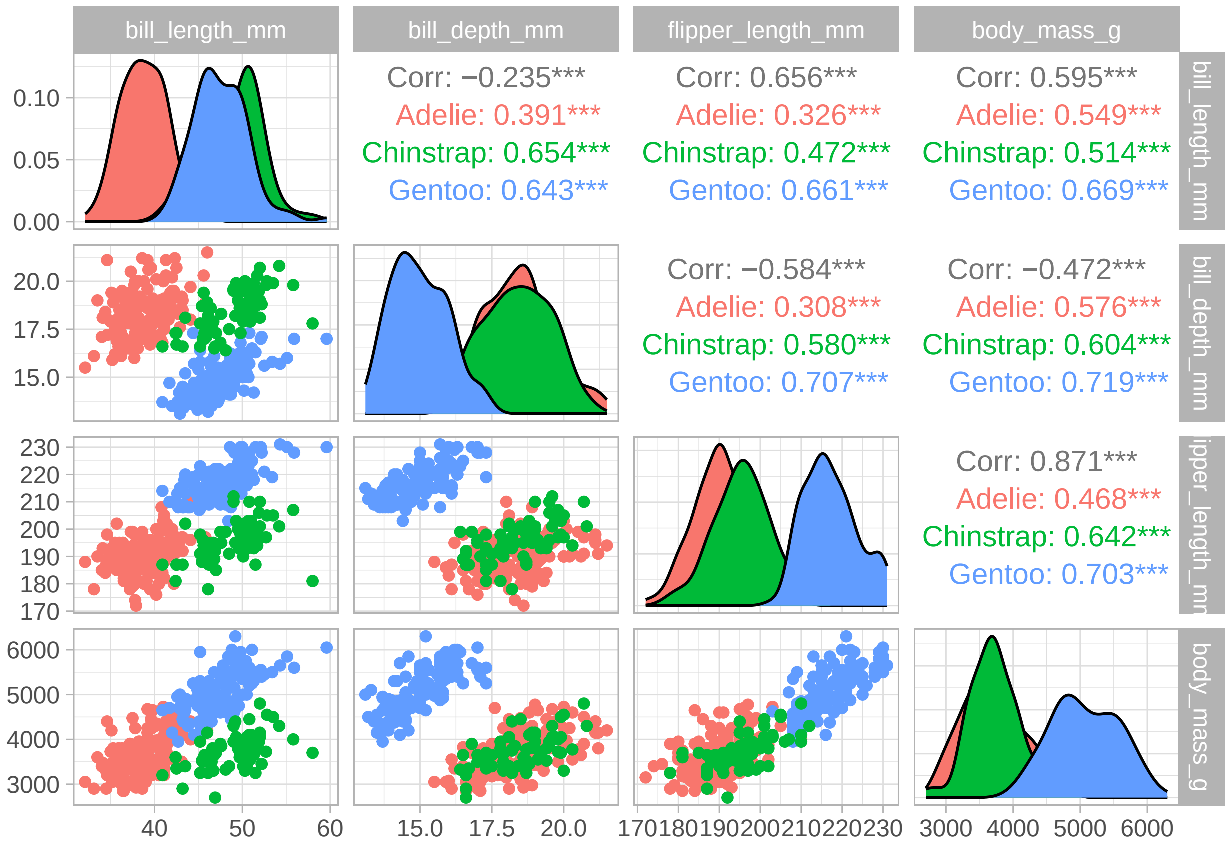

```
###Componentes

peng_pca %>%
  tidy(id = "pca") %>%
  mutate(
    terms = tidytext::reorder_within(
      terms,
      abs(value),
      component
    )
  ) %>%
  ggplot(aes(abs(value), terms, fill = value > 0)) +
  geom_col() +
  facet_wrap(~component, scales = "free_y") +
  tidytext::scale_y_reordered() +
  scale_fill_manual(values = c("#b6dfe2", "#0A537D")) +
  labs(
    x = "Absolute value of contribution",
    y = NULL,
    fill = "¿Positive?"
  )
```




```
peng_pca %>%
  tidy(id = "pca") %>%
  filter(component %in% paste0("PC", 1:3)) %>%
  mutate(component = fct_inorder(component)) %>%
  ggplot(aes(value, terms, fill = terms)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~component, nrow = 1) +
  labs(y = NULL)
```

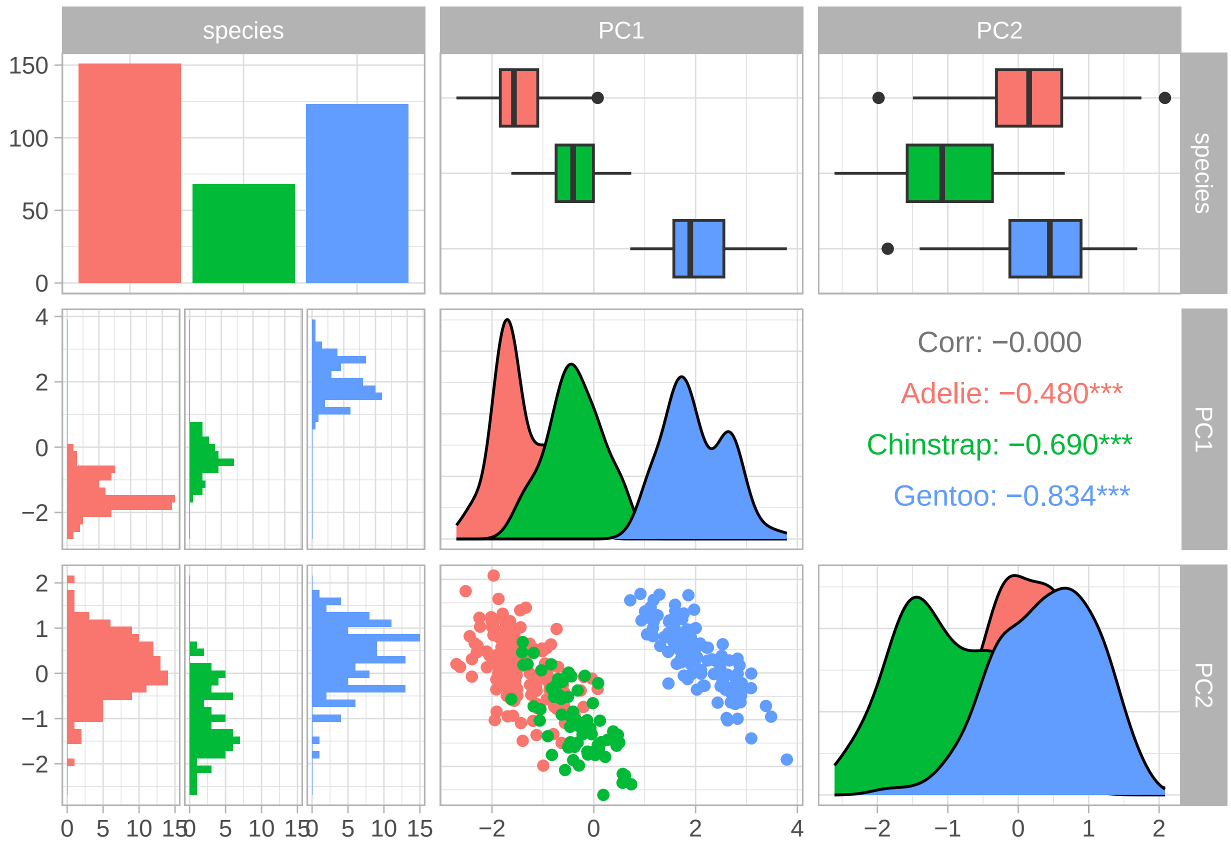



```
###EXTRAS

library(GGally)
library(FactoMineR)
library(factoextra)

df1_con_clase$species <- as.factor(df1_con_clase$species)
datos_pca$species     <- as.factor(datos_pca$species)


ggpairs(
  df1_con_clase,
  mapping = aes(color = species),
  columns = 2:ncol(df1_con_clase)
)
```

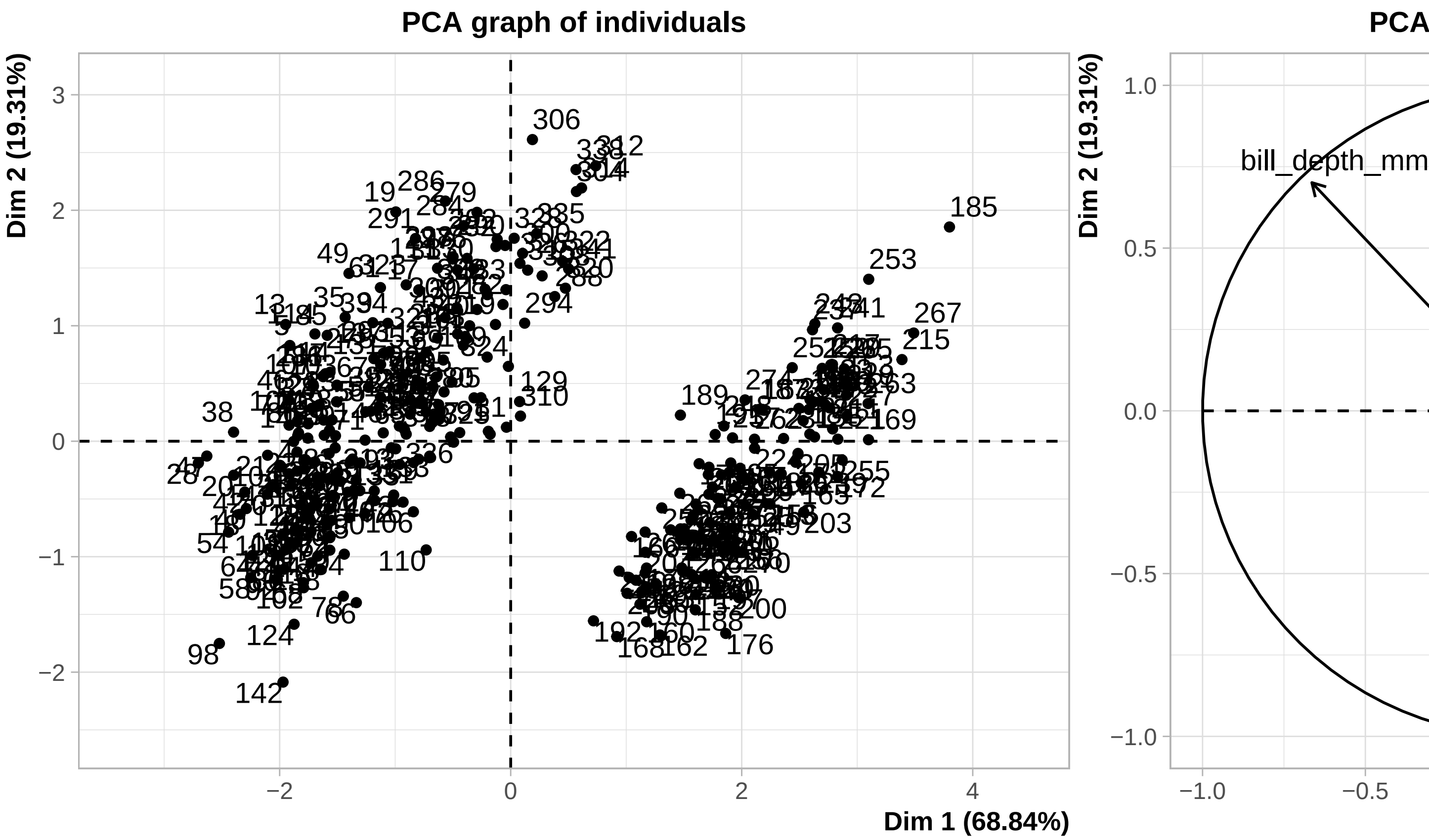



```r
ggpairs(
  datos_pca,
  mapping = aes(color = species),
  columns = 1:3    # PC1, PC2, PC3
)
```

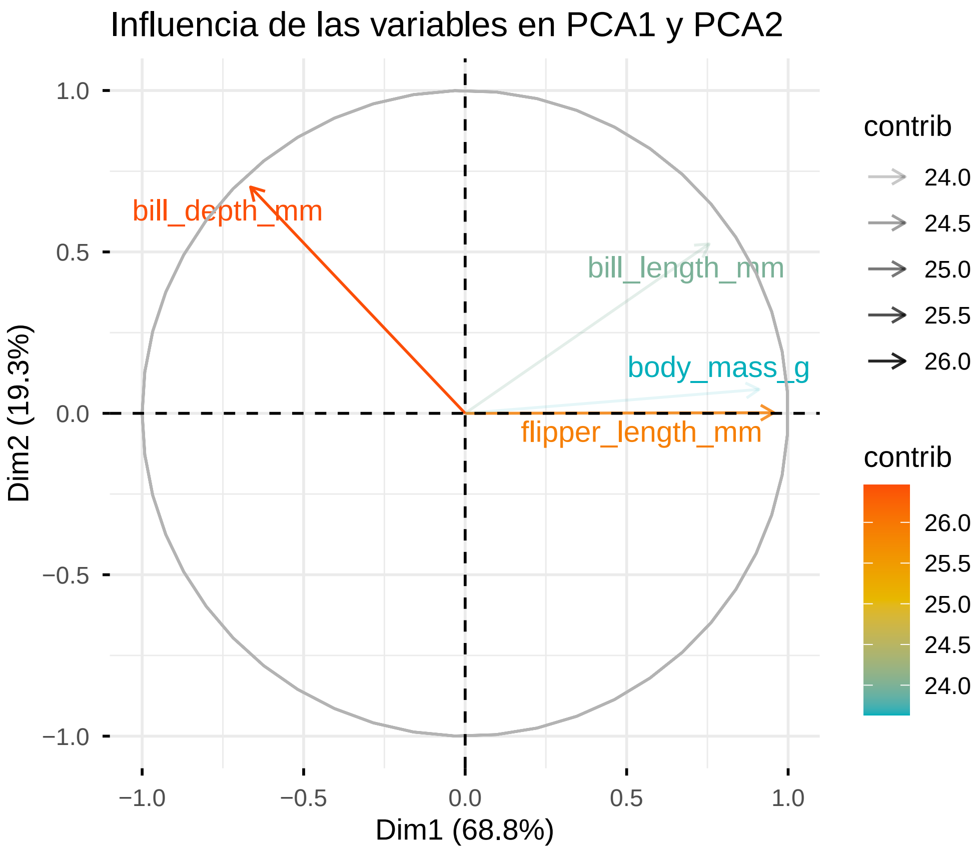
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



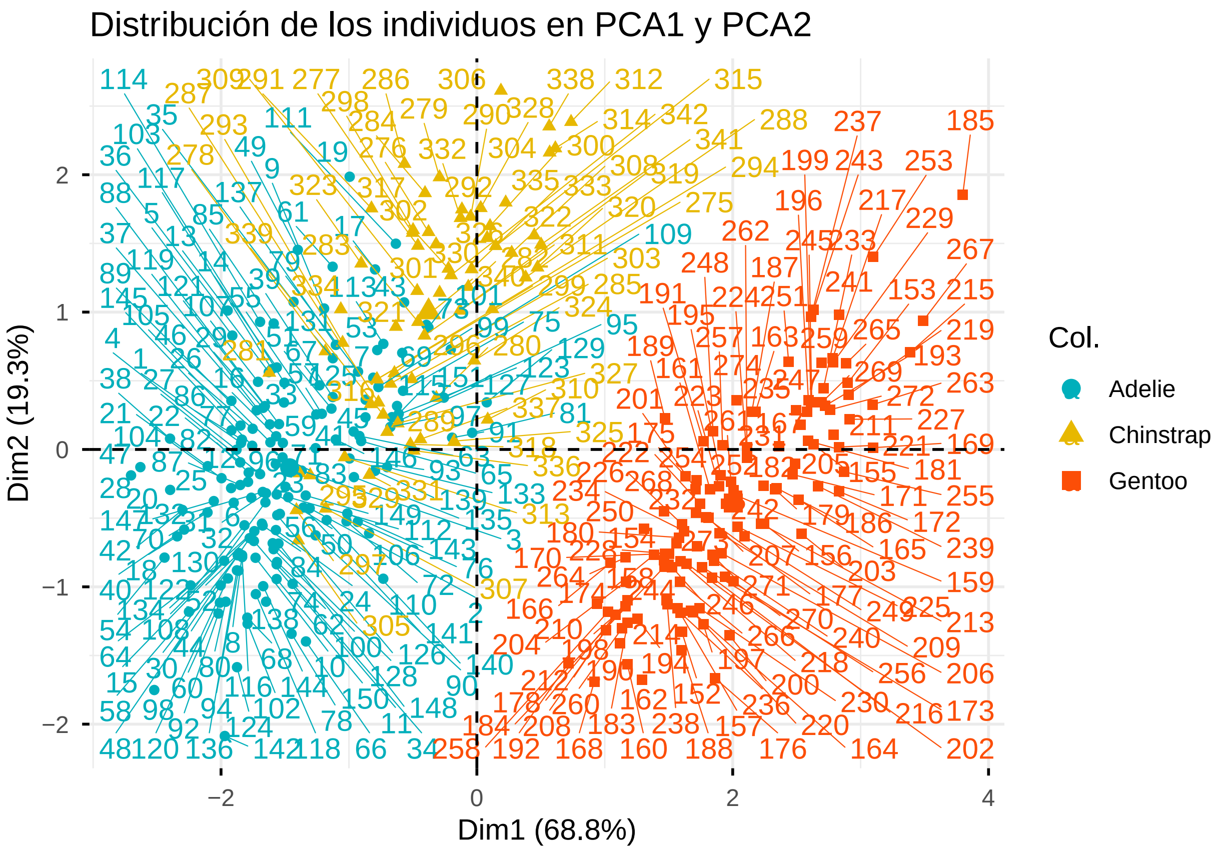

```r
res.pca <- PCA(df1_con_clase[, -1], scale.unit = TRUE)
```



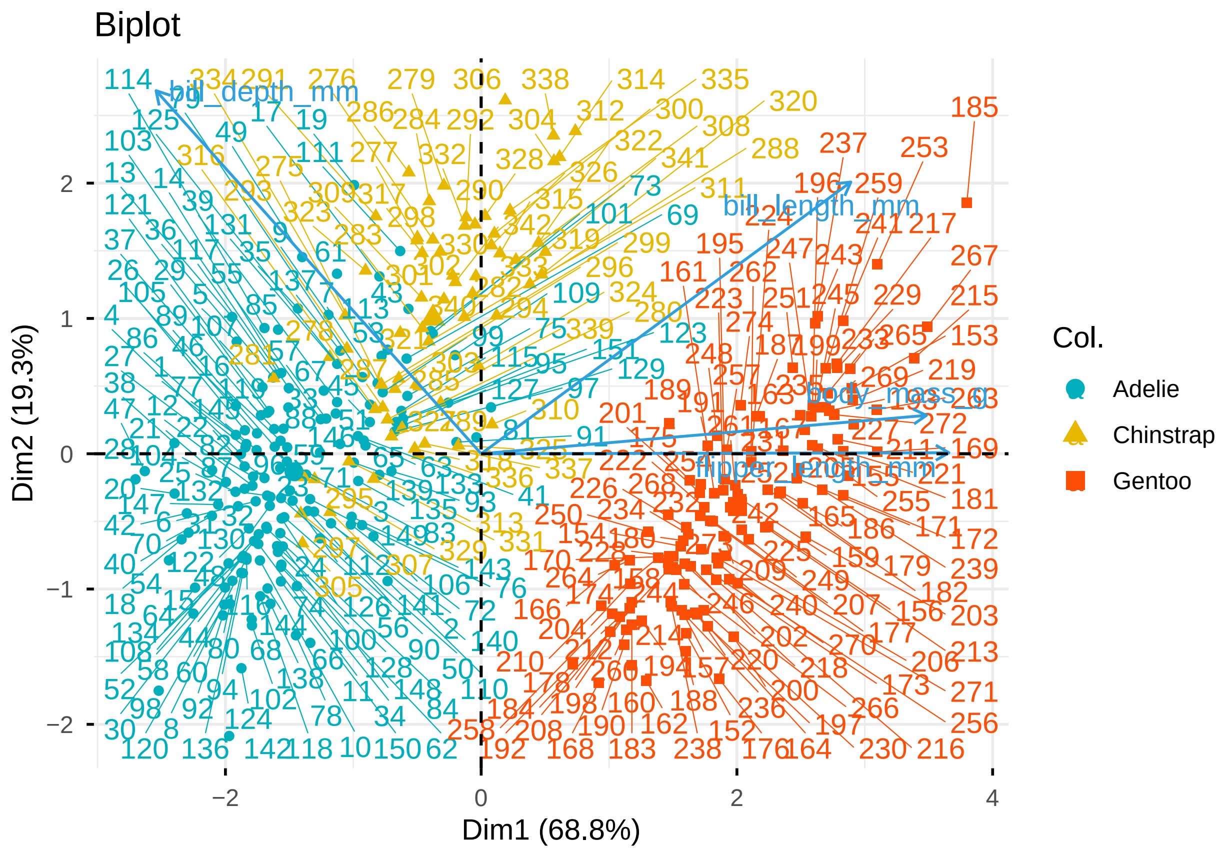

```
fviz_pca_var(
  res.pca,
  alpha.var = "contrib",
  col.var = "contrib",
  gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
  title = "Influencia de las variables en PCA1 y PCA2",
  repel = TRUE
)
```



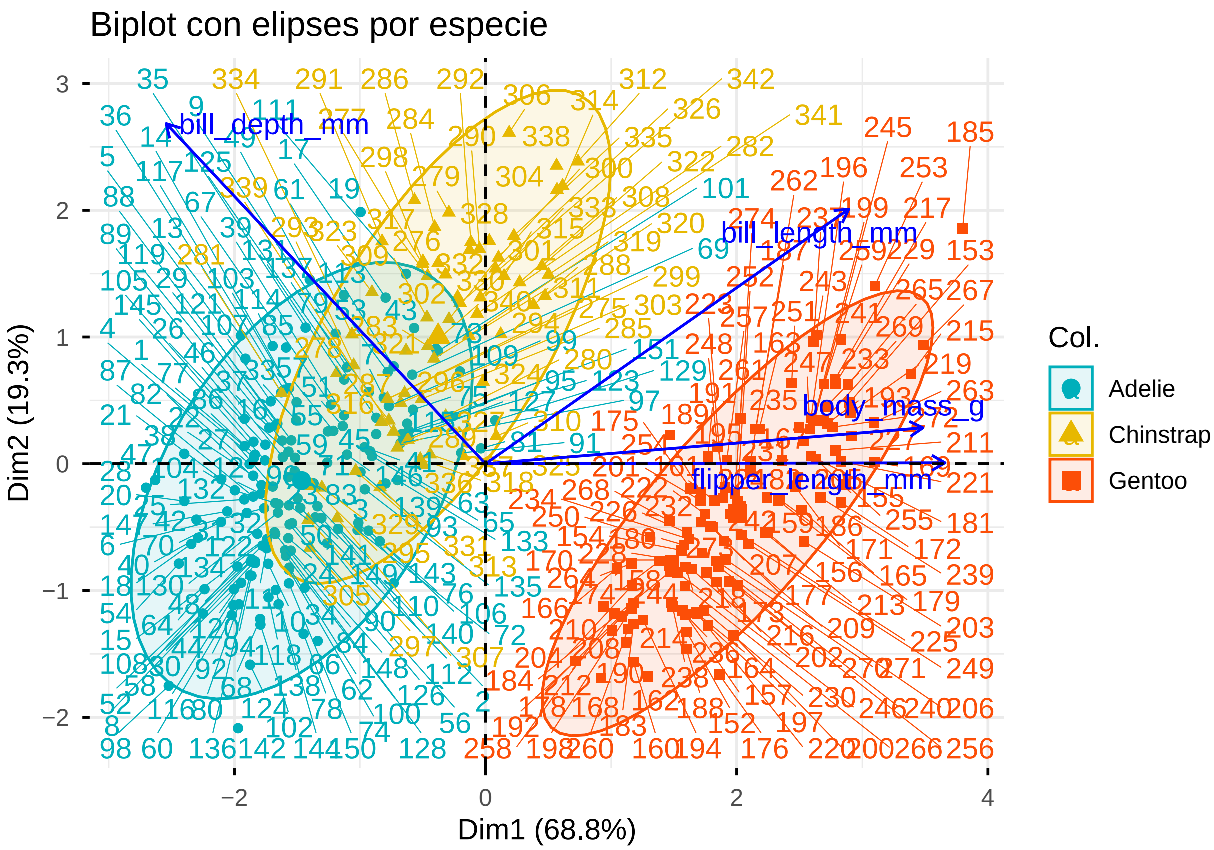

```r
fviz_pca_ind(
  res.pca,
  col.ind = df1_con_clase$species,
  palette = c("#00AFBB", "#E7B800", "#FC4E07"),
  title = "Distribución de los individuos en PCA1 y PCA2",
  repel = TRUE
)
```




```r
fviz_pca_biplot(
  res.pca,
  repel = TRUE,
  title = "Biplot",
  col.var = "#2E9FDF",
  col.ind = df1_con_clase$species,
  palette = c("#00AFBB", "#E7B800", "#FC4E07")
)
```




```r
fviz_pca_biplot(
  res.pca,
  repel = TRUE,
  col.var = "blue",
  col.ind = df1_con_clase$species,
  palette = c("#00AFBB", "#E7B800", "#FC4E07"),
  addEllipses = TRUE,
  ellipse.level = 0.95,
  title = "Biplot con elipses por especie"
)
```


Biplot con elipses por especie
bill_depth_mm
bill_length_mm
body_mass_g
flipper_length_mm
Dim1 (68.8%)
Dim2 (19.3%)
Col.
Adelie
Chinstrap
Gentoo